

GATE 2025
AGRICULTURAL ENGINEERING (AG)

Answers to selected questions

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Q.6 – Q.10 Carry TWO marks Each

Q.8 A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

- (A) $\frac{75}{216}$
 (B) $\frac{1}{6}$
 (C) $\frac{1}{18}$
 (D) $\frac{25}{216}$

Solution

Step 1: Finding the probability mass function(PMF) of a single dice roll:

The sample space is $S \in \{1, 2, 3, 4, 5, 6\}$

Let us define a variable Y_1 as follows:

$$Y_1 = \begin{cases} 1 & X_1 = 6 \\ 0 & X_1 \neq 6 \end{cases}$$

Now, The PMF is defined as:

$$P_{Y_1}(k) = \begin{cases} \frac{1}{6} & k = 1 \\ \frac{5}{6} & k = 0 \end{cases}$$

Let p,q be $\frac{1}{6}$ and $\frac{5}{6}$ respectively.

Step 2: Applying Z-Transform to find the required PMF for Y

Let us define Y_2 and Y_3 as the random variables for the 2nd and 3rd dice roll, respectively. Let Y denote the random variable for the combined triple dice roll.

Now, we can write $Y = Y_1 + Y_2 + Y_3$.

Writing the Z-Transform for Y_1

$$M_{Y_1}(Z) = E[Z^{-Y_1}] = \sum P_{Y_1}(k) Z^{-k} \quad (1)$$

$$\begin{aligned} &= (P_{Y_1}(1)Z^{-1} + P_{Y_1}(0)Z^{-0}) \\ &= pZ^{-1} + q \end{aligned} \quad (2)$$

We know that Y_1, Y_2, Y_3 are independent events. So, we can write $M_Y(Z) = M_{Y_1}(Z) M_{Y_2}(Z) M_{Y_3}(Z)$. Therefore,

$$M_Y(Z) = (pZ^{-1} + q)^3. \quad (\text{From 2})$$

$$\begin{aligned} &= \sum_{k=0}^3 {}^3C_k q^{3-k} (pZ^{-1})^k \\ &= \sum P_Y(k) Z^{-k} \end{aligned} \quad (\text{From 1})$$

So, we can say:

$$P_Y(k) = {}^3C_k q^{3-k} p^k \quad (3)$$

The above situation is called as a binomial distribution.

Step 3: Using binomial distribution to get required probability We need the situation where p appears exactly once. So, we

consider only the term where the power of p is 1.

The required term is : ${}^3C_1 p q^2$.

Substituting the values of p and q, we get:

$$\begin{aligned} & {}^3C_1 \frac{1}{6} \left(\frac{5}{6}\right)^2 \\ &= \frac{75}{216} \end{aligned}$$

Therefore, the required answer is $\frac{75}{216}$

C Implementation

```
#include <stdio.h>
#include<stdlib.h>
#include <time.h>
void roll(int*b);
int main(void){
    int hit=0;
    int fail=0;
    srand(time(NULL));
    int a[]={0,0,0};
    for (int i=0;i<1000000000;i++){
        roll(a);
        if ((a[0]==6 && a[1]!=6 && a[2]!=6) || (a[0]!=6 && a[1]!=6 && a[2]==6) || (a[0]!=6 && a[1]==6 && a[2]!=6))
            //printf("hit");
            hit+=1;
        //printf("%d %d %d",a[0],a[1],a[2]);
    }
    else{
        fail+=1;
    }
    }
    printf("hits %d, Fail %d\n",hit,fail);
    printf("\nratio=%lf\n",(double)(hit)/(hit+fail));

    return 0;
}
void roll(int*b){
    b[0]=rand()%6+1;
    b[1]=rand()%6+1;
    b[2]=rand()%6+1;
}
```

The terminal output is shown below:

```
madhur@madhur-Dell-15-DC15250:~$ gcc test.c
madhur@madhur-Dell-15-DC15250:~$ ./a.out
hits 347221618, Fail 652778382

ratio=0.347222
```

The answer $\left(\frac{75}{216}\right)$ is 0.34722222.. so, we can say that our answer is correct.

Q.11 – Q.35 Carry ONE mark Each

Q.11 If a, b, and c are the roots of $2x^3 - 3x^2 + px - 1 = 0$ and sum of the two roots is 1, the value of p is:

- (A) 2
- (B) 1
- (C) 3
- (D) $\frac{1}{3}$

Solution

Let the roots whose sum is 1 be a and b.

Then, by using the properties of coefficients of an equation, we write:

$$\begin{aligned} -\left(\frac{-3}{2}\right) &= a + b + c \\ \frac{3}{2} &= 1 + c \\ c &= \frac{1}{2} \end{aligned} \tag{1}$$

And

$$\begin{aligned} abc &= -\left(\frac{-1}{2}\right) \\ ab \frac{1}{2} &= \frac{1}{2} \\ ab &= 1 \end{aligned} \tag{From (1)}$$

Also,

$$\begin{aligned} ab + bc + ac &= \left(\frac{p}{2}\right) \\ 1 + c(a + b) &= \frac{p}{2} \\ 1 + \frac{1}{2}(1) &= \frac{p}{2} \\ 3 &= p \end{aligned} \tag{From (2)}$$

Therefore, the value of p is 3.

Q.12 If the matrix $\begin{bmatrix} x & -3 \\ 2 & x-5 \end{bmatrix}$ is singular, sum of the values of x is:

- (A) 5
- (B) 6
- (C) 0
- (D) -1

Solution

A matrix is said to be singular if its row echelon form has a row full of zeroes.

Step 1: Finding row echelon form of the given matrix:

$$\begin{bmatrix} x & -3 \\ 2 & x-5 \end{bmatrix}$$

$$\begin{bmatrix} x & -3 \\ 1 & \frac{(x-5)}{2} \end{bmatrix}$$

$$\begin{bmatrix} x - [(x)1] & -3 - [(x)\frac{(x-5)}{2}] \\ 1 & \frac{(x-5)}{2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & \frac{(-6-x^2+5x)}{2} \\ 1 & \frac{(x-5)}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{(x-5)}{2} \\ 0 & \frac{(-6-x^2+5x)}{2} \end{bmatrix}$$

Step 2: Making the bottom row to be full of zeroes:

We need to equate $\frac{(-6-x^2+5x)}{2}$ to zero.

We get

$$-6 - x^2 + 5x = 0$$

$$x^2 - 5x + 6 = 0$$

So, the sum of values of x is 5.

Since it is not possible to display 'x' and 'x-5' explicitly in the code, a numerical approach was taken:

The following code converts a numerical matrix into its row echelon form.

We can verify that the values of $(x = 2, 3)$ give us a matrix with a row of zeroes

C Implementation

```
#include <stdio.h>
#include <math.h>
#define ROWS 2
#define COLS 2

void printmtrx(double mtrx[ROWS][COLS]);
void convertToEchelon(double mtrx[ROWS][COLS]);

int main() {

    double mtrxa[ROWS][COLS] = { //for x=2
        { 2, 2},
        {-3, -3}

    };

    double mtrx[ROWS][COLS] = { //for x=3
        {3, -3},
        {2, -2}
```

```

};
printf("Original matrix A :\n");
printmtrx(mtrxa);
printf("Original matrix B :\n");
printmtrx(mtrxb);
convertToEchelon(mtrxa);
convertToEchelon(mtrxb);

printf("Echelon Form a:\n");
printmtrx(mtrxa);
printf("\n Echelon Form b : \n");
printmtrx(mtrxb);
return 0;
}

void printmtrx(double mtrx[ROWS][COLS]) {
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            printf("%0.2f ", mtrx[i][j]);
        }
        printf("\n");
    }
    printf("\n");
}

void convertToEchelon(double mtrx[ROWS][COLS]) {
    int lead = 0; // Tracks the current pivot column
    for (int r = 0; r < ROWS; r++) {
        if (lead >= COLS) {
            return;
        }
        // Finding non zero element in column
        int i = r;
        while (mtrx[i][lead] == 0) {
            i++;
            if (i == ROWS) {
                i = r;
                lead++;
                if (lead == COLS) {
                    return;
                }
            }
        }
        // Swap current row 'r' with pivot row 'i'
        if (i != r) {
            for (int j = 0; j < COLS; j++) {
                double temp = mtrx[r][j];
                mtrx[r][j] = mtrx[i][j];
                mtrx[i][j] = temp;
            }
        }
        // 2. Eliminate entries below the current pivot
        for (int k = r + 1; k < ROWS; k++) {
            if (fabs(mtrx[k][lead]) > 1e-9) {
                double factor = mtrx[k][lead] / mtrx[r][lead];
                for (int j = lead; j < COLS; j++) {

```

```

        mtrx[k][j] -= factor * mtrx[r][j];
    }
}
}

lead++;
}
}

```

The terminal output is :

```

madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ gcc echelon.c -lm
madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ ./a.out
Original matrix A :
2.00 2.00
-3.00 -3.00

Original matrix B :
3.00 -3.00
2.00 -2.00

Echelon Form a:
2.00 2.00
0.00 0.00

Echelon Form b :
3.00 -3.00
0.00 0.00

```

Since both gave a row echelon form with a row of zeroes, we can conclude that $x = 2, 3$ are the 2 values satisfying the singular condition. So, their sum is 5.

Q.13 $\frac{s}{s^2 + a^2}$ is the Laplace transform of:

- (A) $\cos(at)$
- (B) $\sin(at)$
- (C) $\sinh(at)$
- (D) $\cosh(at)$

Solution

We do not know the laplace transforms of the options, so we hit-and-trial them to get the expression:

Option (A):

$$\begin{aligned}
\mathcal{L}\{\cos(at)\} &= \int_0^\infty \cos(at) e^{-st} dt = I \\
I &= \int_0^\infty \frac{d(\sin(at))}{a} e^{-st} \\
&= \frac{\sin(at)}{a} e^{-st} \Big|_0^\infty - \frac{1}{a} \int_0^\infty \sin(at) e^{-st} (-s) dt \\
&= 0 + \frac{s}{a} \int_0^\infty \frac{d(-\cos(at))}{a} e^{-st} \\
&= -\left(\frac{s}{a^2}\right) \left\{ \cos(at) e^{-st} \Big|_0^\infty - \int_0^\infty \cos(at) e^{-st} (-s) dt \right\} \\
&= -\left(\frac{s}{a^2}\right) \{(0 - 1) - (-s)I\} \\
&= -\left(\frac{s}{a^2}\right) \{-1 + sI\} \\
I &= \frac{s}{a^2} - \frac{s^2}{a^2}(I) \\
\left(\frac{a^2 + s^2}{a^2}\right) I &= \frac{s}{a^2} \\
I &= \frac{s}{s^2 + a^2}
\end{aligned}$$

(continued in the next page)

Option (B):

$$\begin{aligned}
\mathcal{L}\{\sin(at)\} &= \int_0^\infty \sin(at) e^{-st} dt = I \\
I &= \int_0^\infty \frac{d(-\cos(at))}{a} e^{-st} \\
&= \frac{-\cos(at)}{a} e^{-st} \Big|_0^\infty - \frac{1}{a} \int_0^\infty -\cos(at) e^{-st} (-s) dt \\
&= \frac{1}{a} - \frac{s}{a} \int_0^\infty \frac{d(\sin(at))}{a} e^{-st} \\
&= \frac{1}{a} - \left(\frac{s}{a^2}\right) \left\{ \sin(at) e^{-st} \Big|_0^\infty - \int_0^\infty \sin(at) e^{-st} (-s) dt \right\} \\
&= -\left(\frac{s}{a^2}\right) \{(0 - 0) - (-s)I\} \\
&= -\left(\frac{s}{a^2}\right) \{0 + sI\} \\
I &= \frac{a}{a^2} - \frac{s^2}{a^2}(I) \\
\left(\frac{a^2 + s^2}{a^2}\right) I &= \frac{a}{a^2} \\
I &= \frac{a}{s^2 + a^2}
\end{aligned}$$

Option (C):

$$\begin{aligned}
 \mathcal{L}\{\sinh(at)\} &= \int_0^\infty \frac{e^{at} - e^{-at}}{2} e^{-st} dt \\
 &= \frac{1}{2} \int_0^\infty e^{(a-s)(t)} dt - \frac{1}{2} \int_0^\infty e^{(-a-s)(t)} dt \\
 &= \frac{1}{2} \left\{ \frac{e^{(a-s)(t)}}{a-s} \right\} \Big|_0^\infty - \frac{1}{2} \left\{ \frac{e^{(-a-s)(t)}}{-a-s} \right\} \Big|_0^\infty \quad (\text{assume } s > a \text{ for integral to be finite}) \\
 &= \frac{1}{2} \left(\frac{1}{s-a} - \frac{1}{s+a} \right) \\
 &= \frac{1}{2} \left(\frac{2a}{s^2 - a^2} \right) \\
 I &= \frac{a}{s^2 - a^2}
 \end{aligned}$$

Option (D):

$$\begin{aligned}
 \mathcal{L}\{\cosh(at)\} &= \int_0^\infty \frac{e^{at} + e^{-at}}{2} e^{-st} dt \\
 &= \frac{1}{2} \int_0^\infty e^{(a-s)(t)} dt + \frac{1}{2} \int_0^\infty e^{(-a-s)(t)} dt \\
 &= \frac{1}{2} \left\{ \frac{e^{(a-s)(t)}}{a-s} \right\} \Big|_0^\infty + \frac{1}{2} \left\{ \frac{e^{(-a-s)(t)}}{-a-s} \right\} \Big|_0^\infty \quad (\text{assume } s > a \text{ for integral to be finite}) \\
 &= \frac{1}{2} \left(\frac{1}{s-a} + \frac{1}{s+a} \right) \\
 &= \frac{1}{2} \left(\frac{2s}{s^2 - a^2} \right) \\
 I &= \frac{s}{s^2 - a^2}
 \end{aligned}$$

The option that has the same laplace transform as the one in the question is Option(A). So, it is the correct answer.

Q.15 The median of the following set of numbers: 154, 130, 144, 137, 156, 146, 138, 149, 160, 138 is:

- (A) 145.0
- (B) 145.2
- (C) 151.0
- (D) 146.0

Solution

To find the median, we need to arrange the data in ascending or descending order.
After arranging them in ascending order, the data is as follows:

{130,137,138,138,144,146,149,154,156,160}

The number of elements in the data set are 10.

Since it is an even number, the median (M) is defined as the average of the middle 2 terms (in this case, average of 5th and 6th terms).

Therefore, the median(M) is $\frac{144 + 146}{2} = 145.0$.

C Implementation

```

#include <stdio.h>

void swap(int*a,int i,int j);
int partition(int*a, int low, int high);
int quicksort(int*a,int low,int high);

int main(){
    int a[]={130,137,138,138,144,146,149,154,156,160};
    int q=sizeof(a)/sizeof(a[0]);
    //Printing original array
    printf("The original array is : [");
    for(int k=0;k<q;k++){
        printf("%d ",a[k]);
    }
    printf("]\n");
    //printf("%d\n", q);
    quicksort(a,0,q-1);
    printf("The sorted array is: [");
    for(int k=0;k<q;k++){
        printf("%d ",a[k]);
    }
    printf("]\n");
    double m;
    if (q%2==0){
        m=(a[(q/2)]+a[(q/2-1)])/2;
        printf("\nThe median is : %lf\n", m);
    }
    else {
        m=(a[(q-1)/2]);
        printf("\nThe median is : %lf\n", m);
    }
}

int partition(int* a,int low, int high){
    int pivot = a[high];
    int i=low-1;
    for (int j=low;j<high;j++){
        if(a[j]<pivot){
            i+=1;
            swap(a,i,j);
        }
    }
    swap(a,i+1,high);
    return i+1;
}

int quicksort(int *a, int low ,int high){
    if(low<high){
        int pi=partition(a,low,high);
        quicksort(a,low,pi-1);
        quicksort(a,pi+1,high);
    }
}

```

```
void swap(int*a,int i,int j){
    int temp=a[i];
    a[i]=a[j];
    a[j]=temp;
}
```

The output is as follows:

```
madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ gcc Sorter.c
madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ ./a.out
The original array is : [130 137 138 138 144 146 149 154 156 160 ]
The sorted array is: [130 137 138 138 144 146 149 154 156 160 ]

The median is : 145.000000
```

Q.36 – Q.65 Carry TWO marks each

Q.36 The series

$\sum_{n=0}^r q^n = 1 + q + q^2 + \dots$ has the sum:

(A) $\frac{1}{1-q} - \frac{q^{r+1}}{1-q}$

(B) $\frac{1}{q-1} + \frac{q^{r+1}}{q-1}$

(C) $\frac{1}{1-q} + \frac{q^{r+1}}{1-q}$

(D) $\frac{1}{q-1}$

Solution

The given sequence is a geometric progression with initial term as 1 and common ratio as q and the number of terms are r+1.

The sum of a geometrical progression with initial term a, common ratio r and number of terms n is given by:

$$a \frac{r^n - 1}{r - 1}$$

applying the above formula, we get :

$$1 \frac{q^{r+1} - 1}{q - 1}$$

$$\frac{1 - q^{r+1}}{1 - q}$$

$$\frac{1}{1 - q} - \frac{q^{r+1}}{1 - q}$$

Therefore, the answer is option A.

C implementation

```
#include<stdio.h>
#include<math.h>
int main(){
    double q,sum=0;
    int r;
    printf("Enter the value of q:");
    scanf("%lf",&q);
    printf("\nEnter the value of r:");
```

```
scanf("%d",&r);
for(int i=0;i<=r;i++){
    sum+=pow(q,i);
}
double formula =(1/(1-q))-((pow(q,r+1))/(1-q));
printf("Sum is %lf\n", sum);
printf("The formula gave %lf\n",formula);
if (sum==formula){
printf("Both are equal\n");
}
}
```

```
madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ gcc gp.c -lm
madhur@madhur-Dell-15-DC15250:~/LATEX_CODES$ ./a.out
Enter the value of q:2

Enter the value of r:3
Sum is 15.000000
The formula gave 15.000000
Both are equal
```

Q.37 The value of $\lim_{x \rightarrow 0} \frac{x \cos(x) - \sin(x)}{x^2 \sin(x)}$

- (A) $-\frac{1}{3}$
- (B) $\frac{1}{3}$
- (C) 3
- (D) -3

Solution

We use the idea of Taylor series expansion to solve the problem.

$$\frac{\frac{x(\cos(x)) - \sin(x)}{x^2 \sin(x)}}{\frac{x(1 - \frac{x^2}{2!} \dots) - (x - \frac{x^3}{3!} \dots)}{x^2(x - \frac{x^3}{3!} \dots)}}$$

When x approaches zero, we can ignore the higher power terms.

$$\frac{(x - \frac{x^3}{2}) - (x - \frac{x^3}{6})}{x^3(1 - \frac{x^2}{6})} = \frac{-\frac{1}{3}}{1 - \frac{x^2}{6}}$$

Substituting x as zero gives us the limit as $-\frac{1}{3}$.

Python Implementation

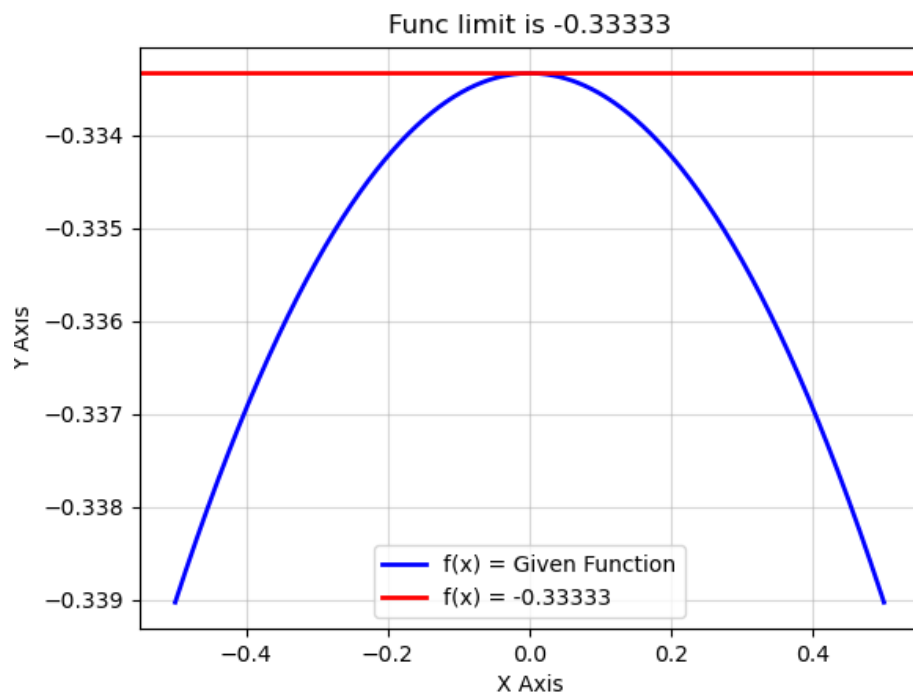
```
import matplotlib.pyplot as plt
import numpy as np

def f(x):
    return (x*(np.cos(x))-np.sin(x))/((x**2)*np.sin(x))

x = np.linspace(-0.5, 0.5, 1000)
y = f(x)
plt.plot(x, y, color='blue', linestyle='-', linewidth=2, label='f(x) = Given Function')
plt.axhline(y=-1/3, color='red', linestyle='-', linewidth=2, label='f(x) = -0.33333')
```

```
plt.title('Func limit is -0.33333')  
plt.xlabel('X Axis')  
plt.ylabel('Y Axis')  
plt.grid(True, alpha=0.5)  
plt.legend()  
plt.show()
```

The output graph is shown below:



Q.38 If $u = x^4 + y^4 + 3x^2y^2$, the value of $\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}\right)^{-1}$

- (A) $4u$
- (B) 0
- (C) $\frac{1}{4u}$
- (D) $2u$

Solution

We can directly evaluate the partial derivative using the chain and product rules, but another approach is to use Euler's theorem for homogeneous functions.

We will solve it using the second method, since the first one is straightforward.

Step 1: Proving the given function is homogeneous:

A function $f(x, y)$ is said to be homogeneous and having degree n if $f(\alpha x, \alpha y) = \alpha^n f(x, y)$

now, let

$$\begin{aligned} u &= g(x, y) = x^4 + y^4 + 3x^2y^2 \\ g(\alpha x, \alpha y) &= \alpha^4(x^4 + y^4 + 3x^2y^2) \end{aligned}$$

Therefore, u is a homogeneous function of degree 4.

Step 2: Using Euler's theorem to get the answer

Euler's Theorem for homogeneous functions

It states that if a function $f(x, y)$ is homogeneous, then the following condition is satisfied:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n f$$

Since we know that the given equation is homogeneous and of degree 4, we can apply this theorem.

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 4u$$

Since they asked for its reciprocal, the answer is $\frac{1}{4u}$

Q.43 The value of y at $x = 0.3$ and $y(x = 0) = 0$ for the differential equation $\frac{dy}{dx} = 3y + 2e^x$ is _____.
(Rounded off to 3 decimal places)

Solution

Step 1: Defining a function $G(s)$ using laplace transform:

$$\begin{aligned} G(s) &= \mathcal{L}\{y\} \\ sG(s) &= \mathcal{L}\{y'\} \end{aligned} \quad (\text{By property of laplace transform and Given } y(0)=0)$$

Step 2: Applying the laplace transform on the differential equation:

$$\mathcal{L}\{y'\} = \mathcal{L}\{3y + 2e^x\}$$

$$\mathcal{L}\{y'\} = 3\mathcal{L}\{y\} + 2\mathcal{L}\{e^x\}$$

$$sG(s) = 3G(s) + 2\mathcal{L}\{e^x\}$$

(Using linearity of laplace transform)

Step 3: Finding laplace transform of e^x :

$$\mathcal{L}\{e^x\} = \int_0^\infty e^t e^{-st} dt$$

$$= \int_0^\infty e^{(1-s)(t)} dt$$

$$= \frac{1}{1-s} (e^{(1-s)(t)} \Big|_0^\infty)$$

$$= \frac{1}{1-s} (0 - 1)$$

$$= \frac{1}{s-1}$$

(Assuming that $s > 1$)Step 4: Finding $G(s)$:

$$G(s)(s-3) = \frac{2}{s-1}$$

$$G(s) = \frac{2}{(s-1)(s-3)}$$

Using partial fractions:

$$G(s) = \left(\frac{1}{s-3} - \frac{1}{s-1} \right)$$

Step 5: Using inverse laplace transform to find y :

$$\mathcal{L}\{y\} = G(s)$$

$$y = \mathcal{L}^{-1} \left\{ \frac{1}{s-3} - \frac{1}{s-1} \right\}$$

$$y = \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\} - \mathcal{L}^{-1} \left\{ \frac{1}{s-1} \right\}$$

(Using linearity of inverse laplace transform)

$$y = e^{3x} - e^x$$

(From above, we say that $\mathcal{L}\{e^{ax}\} = \frac{1}{s-a}$ provided $s > a$.)Step 6: Substituting condition to find value of y :

$$x = 0.3$$

(given)

$$y = e^{3 \times 0.3} - e^{0.3}$$

$$y = 1.110$$

Python Implementation

```
# The expression is y=e^{3x}-e^{x}
#evaluating the expression:
import math
x=0.3
y=math.exp(3*x)-math.exp(x)
print(y)
```

Output :

1.1097443035809462